Efficiency comparison across LLM, human, and non-LLM ARC analyses

Codex analysis

April 7, 2026

Abstract

We compare human, LLM, and non-LLM efficiency signals on ARC-derived data. The strongest cross-source signal is shared task difficulty: human solve rate and mean LLM score correlate moderately, while effort alignment is much weaker. Within LLMs, more thinking budget tracks better performance but also more cost and duration. Within humans, higher ability corresponds to better performance and shorter times. These differences suggest a shared difficulty axis plus source-specific efficiency regimes.

1 Data

The analysis covers the LLM run logs, human session summaries, public-eval task pairs, and the non-LLM Compress/VARC/TRM runs.

- llm / model_runs: 29 rows; performance metric task_solve_rate / pair_accuracy; effort metric avg_duration_per_task.
- llm / ARC-2 task rollup: 2736 rows; performance metric task_score; effort metric task_duration_seconds.
- human / sessions: 509 rows; performance metric solve_rate; effort metric mean_duration_seconds.
- human / public_eval task pairs: 161 rows; performance metric solve_rate; effort metric mean_duration_seconds.
- non_llm / Compress ARC: 1 rows; performance metric final pick pass2; effort metric iterations.
- non_llm / VARC: 4 rows; performance metric pass@1..4 / oracle; effort metric candidate count / attempt dirs.
- non_llm / TRM progression steps: 10 rows; performance metric pair accuracy / kaggle score; effort metric step.

2 Hypotheses

- Shared ARC task difficulty should align across humans, LLMs, and TRM.
- Better performance should trade off against more resource use, but the sign of that tradeoff may differ by source.
- A small number of latent axes should summarize the efficiency structure.
- Generic efficiency statistics should still identify the source family.

Table 1: Data inventory for the efficiency analysis.

source_family	subtype	rows	performance_metric	primary_effort_metric
llm	model_runs	29	task_solve_rate / pair_accuracy	avg_duration_per_task
llm	ARC-2 task rollup	2736	task_score	task_duration_seconds
human	sessions	509	solve_rate	mean_duration_seconds
human	public_eval task pairs	161	solve_rate	mean_duration_seconds
non_llm	Compress ARC	1	final pick pass2	iterations
non_llm	VARC	4	pass@1..4 / oracle	candidate count / attempt dir
non_llm	TRM progression steps	10	pair accuracy / kaggle score	step

3 Cross-source alignment

There is no direct person-level latent correlation between model theta and human ability in the current data structure, so the shared comparison is task-level alignment.

Table 2: Cross-source alignment on shared ARC-2 and public-eval overlaps.

analysis	group	metric_x	metric_y
Shared-task human vs LLM score	shared_arc2	human_solve_rate	llm_mea
Shared-task human vs LLM duration	shared_arc2	human_solve_rate	llm_mea
Shared-task human vs LLM cost	shared_arc2	human_solve_rate	llm_mea
Shared-task human vs TRM score	shared_arc2	human_solve_rate	trm_bes
Shared-task human vs LLM duration	shared_arc2	human_mean_duration_seconds	llm_mea
Public-eval human vs average model	human_public_eval_overlap	solve_rate	lm_mear
Public-eval human vs best single model	human_public_eval_overlap	solve_rate	lm_best_

4 Within-source efficiency

Table 3: Within-source efficiency correlations.

analysis	group	metric_x	metric_y	n	spe
LLM performance vs thinking rank	llm	performance_rate	thinking_rank	22	¡0.0
LLM duration vs thinking rank	llm	avg_duration_per_task	thinking_rank	22	¡0.0
LLM performance vs duration	llm	performance_rate	avg_duration_per_task	29	¡0.0
LLM duration vs cost	llm	avg_duration_per_task	avg_cost_per_task	29	¡0.0
Human performance vs ability	human	performance_rate	ability	509	¡0.0
Human performance vs duration	human	performance_rate	mean_duration_seconds	509	¡0.0
Human performance vs outfit	human	performance_rate	outfit	509	¡0.0
TRM performance vs step	non_llm/trm	performance_rate	primary_effort	10	¡0.0
TRM performance vs oracle	non_llm/trm	performance_rate	oracle_rate	10	¡0.0

Table 4: Cross-validated regression summaries for task-level prediction.

target	model	n	r2	mae	pearson_r	spearman_r
human_mean_duration_seconds	geometry_only	115	-0.183	96.941	0.009	0.074
human_mean_duration_seconds	geometry_plus_llm_perf	115	-0.260	97.795	0.025	0.109
human_mean_duration_seconds	geometry_plus_llm_perf_effort	115	-0.480	105.045	-0.101	-0.051
human_mean_duration_seconds	geometry_plus_llm_plus_trm	115	-0.499	105.820	-0.103	-0.042
human_solve_rate	geometry_plus_llm_perf	115	0.093	0.177	0.370	0.357
human_solve_rate	geometry_plus_llm_perf_effort	115	0.071	0.180	0.366	0.353
human_solve_rate	geometry_plus_llm_plus_trm	115	0.070	0.179	0.365	0.349
human_solve_rate	geometry_only	115	-0.142	0.200	-0.029	0.010

Table 5: Nested OLS comparisons for the key task-level models.

target	base_model	full_model	n	delta_r2	f
human_solve_rate	geometry_only	geometry_plus_llm_perf	97	0.115	6
human_solve_rate	geometry_plus_llm_perf	geometry_plus_llm_perf_effort	97	0.059	1
human_solve_rate	geometry_plus_llm_perf_effort	geometry_plus_llm_plus_trm	97	0.007	0
human_mean_duration_seconds	geometry_only	geometry_plus_llm_perf	97	0.003	0
human_mean_duration_seconds	geometry_plus_llm_perf	geometry_plus_llm_perf_effort	97	0.072	1
human_mean_duration_seconds	geometry_plus_llm_perf_effort	geometry_plus_llm_plus_trm	97	0.006	0

5 Predictive models

6 Latent structure

Source-specific PCA shows a common efficiency axis within each family. Humans load positively on performance and ability, negatively on time and submissions. LLMs load positively on performance, duration, and cost, which suggests more capability comes with more compute consumption. Non-LLM TRM loads on performance and oracle rate, opposite primary effort. The shared-task

Table 6: Source-specific PCA summaries.

source	pc1_variance	pc2_variance	top_pc1_feature	top_pc1_loading	second_pc1_feature	second_p
llm	0.390	0.330	avg_duration_per_task	0.607	performance_rate	0.538
human	0.492	0.208	performance_rate	0.571	ability	0.537
non_llm	0.610	0.293	oracle_rate	0.515	secondary_effort	-0.503

PCA is more revealing: PC1 is mostly a compute-burden axis driven by LLM duration, cost, and tokens, while PC2 is a human-efficiency axis with human solve rate positive and human duration/submissions negative.

- human_solve_rate: PC1=-0.289, PC2=0.518.
- human_mean_duration_seconds: PC1=0.176, PC2=-0.498.
- human_mean_submissions: PC1=-0.021, PC2=-0.493.
- llm_mean_score: PC1=-0.372, PC2=0.253.

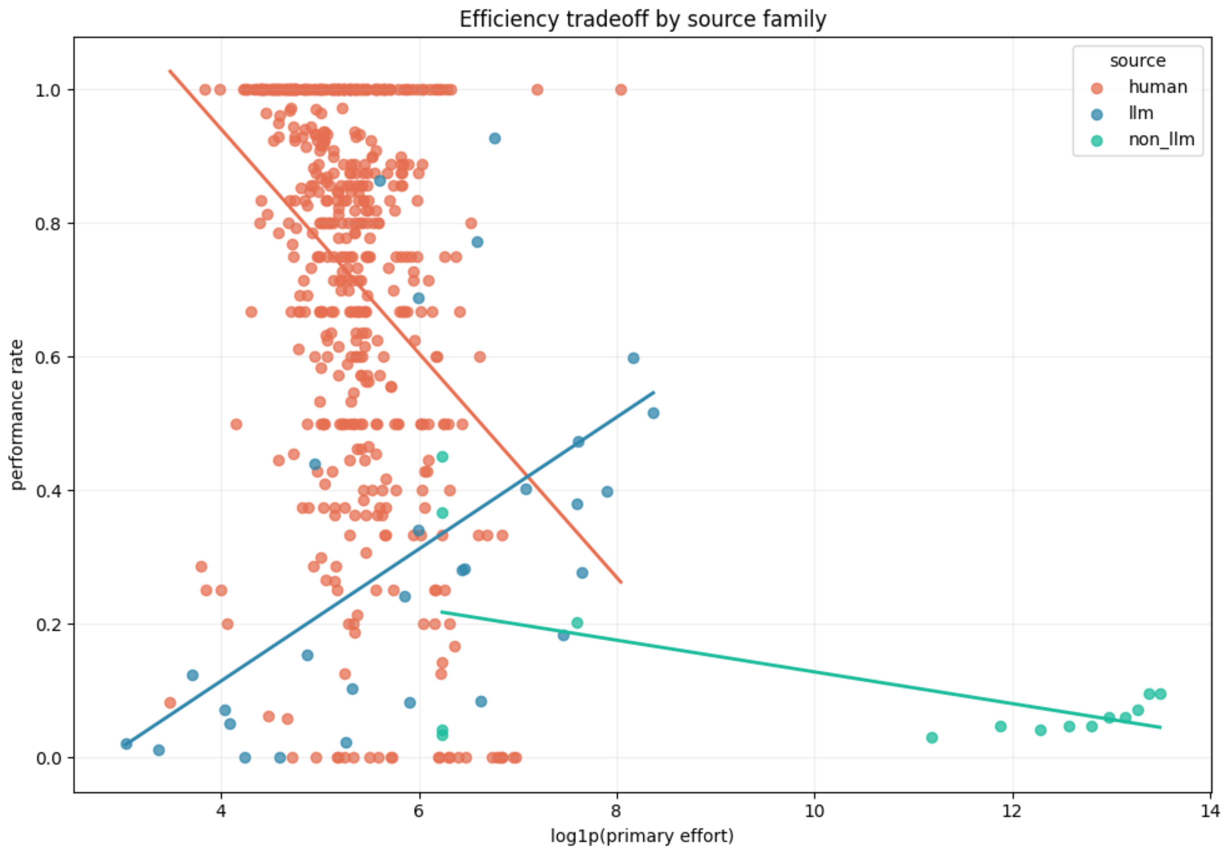
Table 7: Shared-task PCA loadings for the efficiency latent space.

feature	pc1	pc2	explained_variance_ratio_pc1	explained_variance_ratio_pc2
human_solve_rate	-0.289	0.518	0.395	0.165
human_mean_duration_seconds	0.176	-0.498	0.395	0.165
human_mean_submissions	-0.021	-0.493	0.395	0.165
llm_mean_score	-0.372	0.253	0.395	0.165
llm_mean_duration_seconds	0.513	0.212	0.395	0.165
llm_mean_cost	0.522	0.196	0.395	0.165
llm_mean_total_tokens	0.452	0.267	0.395	0.165
trm_best_task_score	-0.073	0.147	0.395	0.165

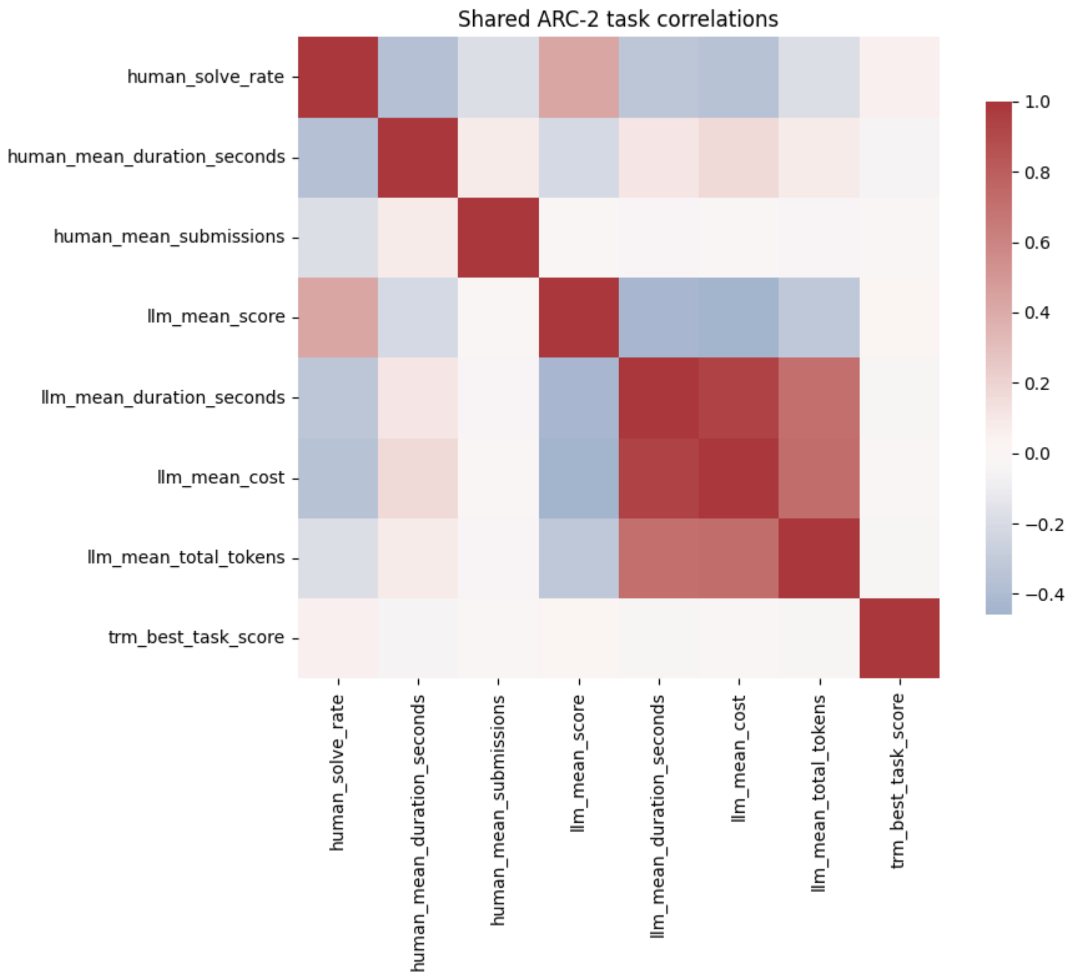
- llm_mean_duration_seconds: PC1=0.513, PC2=0.212.
- llm_mean_cost: PC1=0.522, PC2=0.196.
- llm_mean_total_tokens: PC1=0.452, PC2=0.267.
- trm_best_task_score: PC1=-0.073, PC2=0.147.

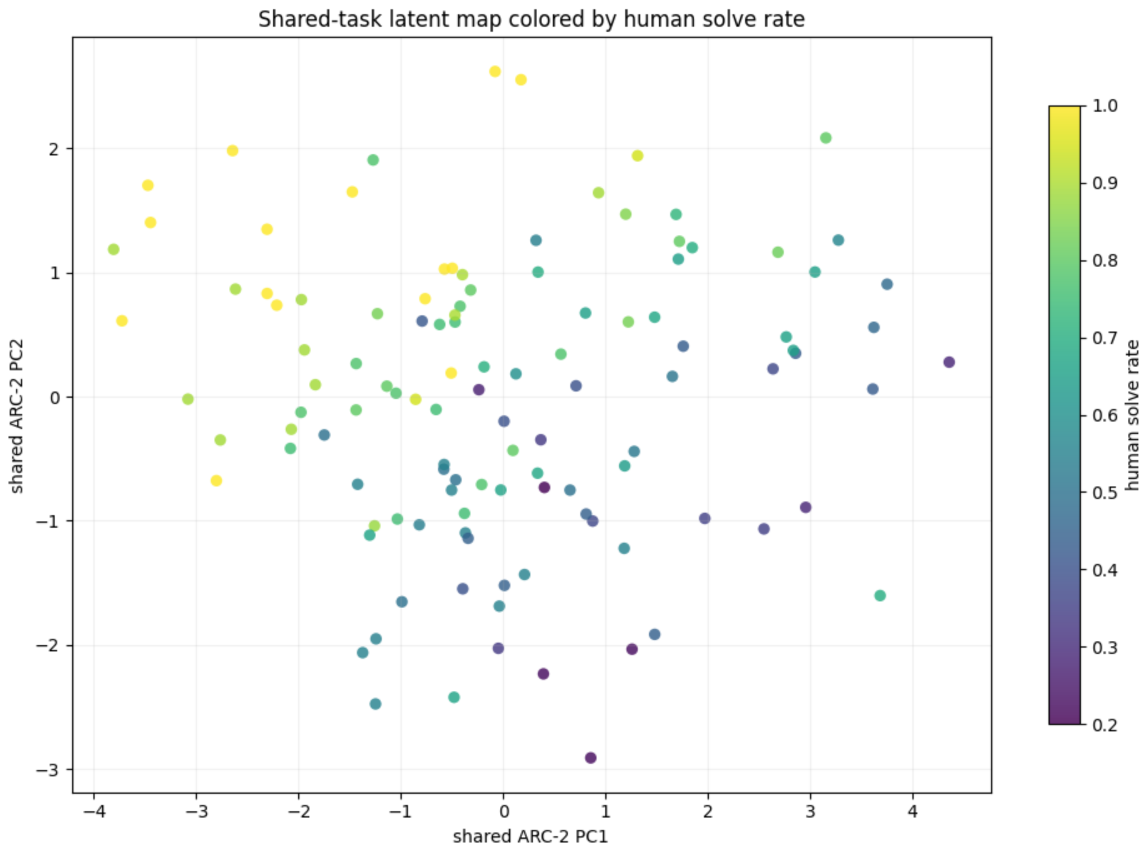
7 Figures

- `figures/source_tradeoff.png`
- `figures/task_correlation_heatmap.png`
- `figures/shared_task_latent_map.png`
- `figures/trm_progression.png`

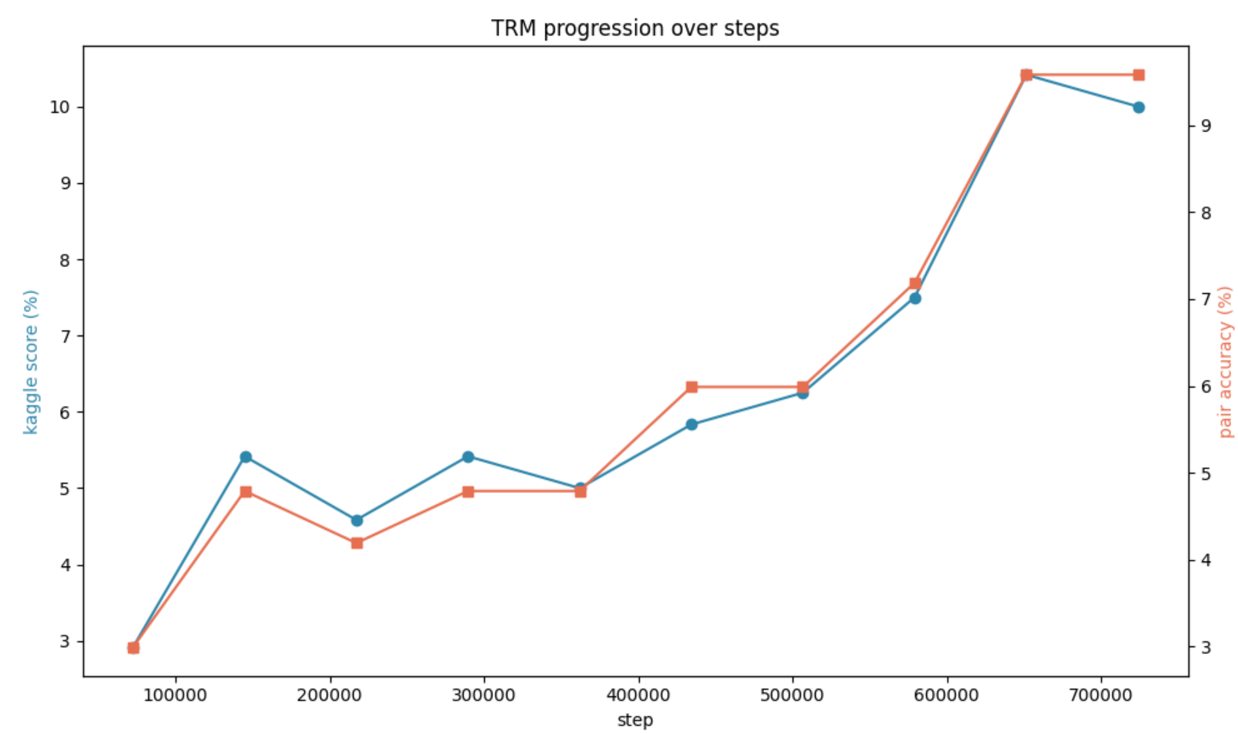


Efficiency tradeoff by source family
source
human
llm
non_llm
performance rate
log1p(primary effort)
0.0
0.2
0.4
0.6
0.8
1.0
4
6
8
10
12
14


Shared ARC-2 task correlations
human_solve_rate
human_mean_duration_seconds
human_mean_submissions
llm_mean_score
llm_mean_duration_seconds
llm_mean_cost
llm_mean_total_tokens
trm_best_task_score
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4



Shared-task latent map colored by human solve rate
shared ARC-2 PC1
shared ARC-2 PC2
human solve rate



TRM progression over steps
step
kaggle score (%)
pair accuracy (%)

8 Non-TRM addendum

This addendum strips TRM out of the non-LLM story and keeps only Compress ARC and VARC. The useful efficiency proxies here are final performance, oracle candidate coverage, selection efficiency, and the pass@k progression. Because Compress ARC has one run and VARC has four, this is descriptive rather than a formal inferential comparison.

8.1 Hypotheses

- Final performance is limited partly by selection, not only candidate generation.
- ARC-1 should be more selection-efficient than ARC-2.
- VARC gains should diminish after pass@2.

8.2 Not pursued

- We did not pool TRM into this addendum, because step count is not on the same scale as candidate-count or iteration proxies.
- We did not treat Compress ARC as inferential evidence, because it contributes only a single run.
- We did not force one unified non-LLM efficiency score across all subtypes, because the telemetry is structurally different across Compress ARC, VARC, and TRM.

8.3 Summary table

Table 8: Non-TRM efficiency summary.

run	performance	oracle	gap	selection	pass1	pass4
Compress ARC	0.203	0.343	0.140	0.591	–	–
ARC-1_Unet	0.367	0.623	0.255	0.590	0.300	0.367
ARC-1_ViT	0.450	0.703	0.253	0.641	0.300	0.450
ARC-2_Unet	0.033	0.092	0.058	0.364	0.017	0.033
ARC-2_ViT	0.042	0.125	0.083	0.333	0.025	0.042

8.4 ARC-1 versus ARC-2 within VARC

Because VARC has two ARC-1 and two ARC-2 runs, we can do an exact two-group permutation check on the split effect. This is still descriptive in spirit, but it gives a clean sanity check for whether ARC-1 is systematically easier to exploit.

8.5 Takeaway

The main pattern is that oracle coverage is consistently larger than final performance, which points to selection as an important bottleneck. Compress ARC and ARC-1 Unet sit around 0.59 selection efficiency, ARC-1 ViT is a bit better at about 0.64, and ARC-2 drops to roughly 0.33–0.36. That makes the ARC-2 VARC runs look less like a search-depth problem and more like a harder candidate-ranking problem.

Table 9: Exact split comparison for the VARC runs.

metric	ARC-1 mean	ARC-2 mean	difference	exact p
performance_rate	0.409	0.037	0.371	0.3333
oracle_rate	0.663	0.108	0.554	0.3333
selection_efficiency	0.615	0.348	0.267	0.3333
pass_gain_1_to_4	0.109	0.017	0.092	0.3333

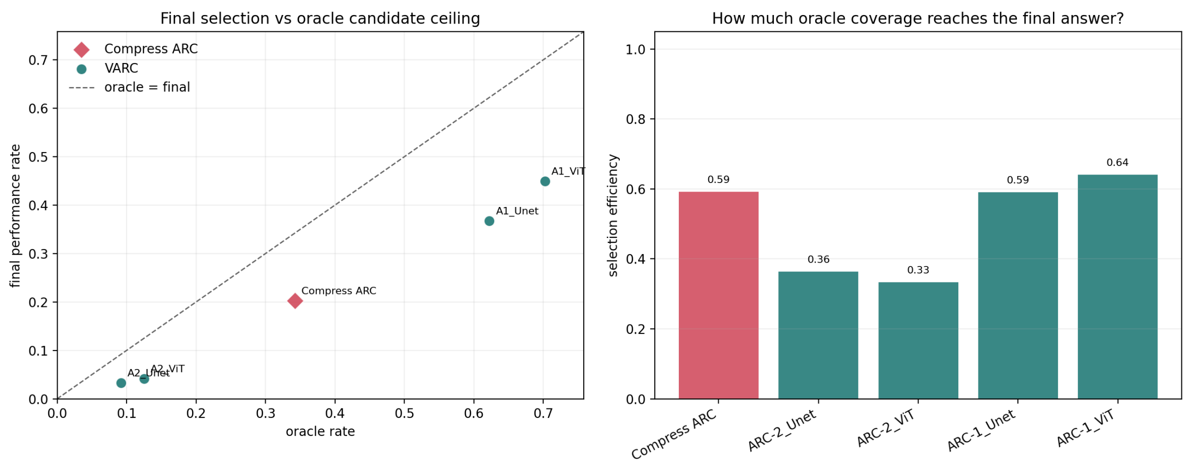


Figure 1: Non-TRM candidate-ceiling view. Left: final performance versus oracle candidate coverage. Right: selection efficiency, defined as final performance divided by oracle rate.

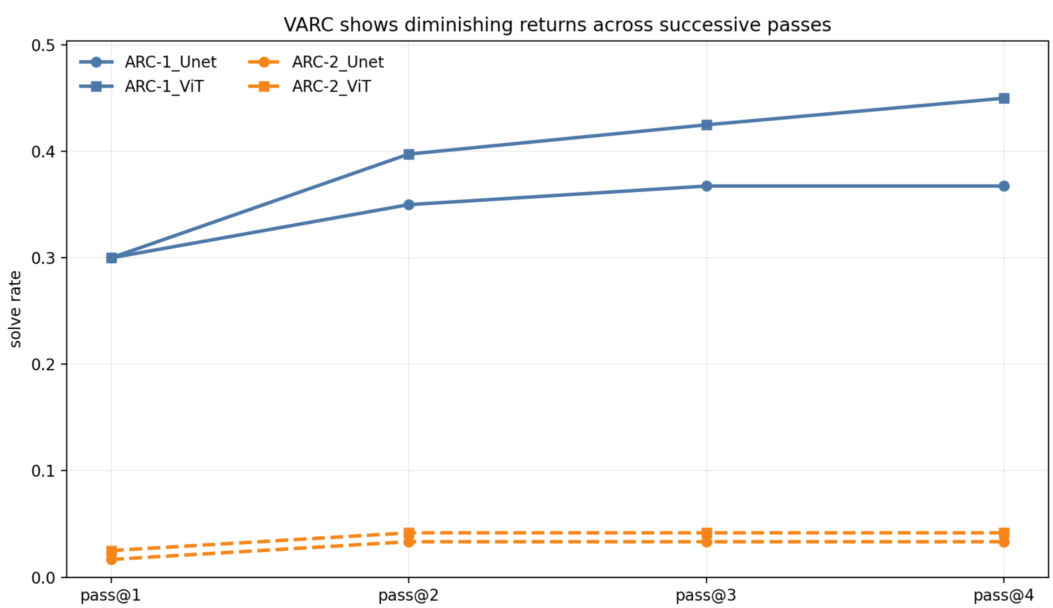


Figure 2: VARC pass@k curves. Gains are front-loaded and flatten quickly, especially on ARC-2.

9 Negative results

- The full-feature source classifier is trivial, because raw telemetry makes the source family obvious.
- Geometry-only prediction of human solve rate is weak.
- Adding TRM after LLM score does not materially improve the shared-task model.
- Human duration is not well predicted by geometry or LLM features.
- Compress ARC has only one row, so it is descriptive only.
- VARC has only four rows, so its correlation structure is fragile.